INTERVENESIM-X

What Should a Robot Learn From a Correction?

A controlled study of correction value, rejected actions, and selective help in simulated manipulation

Matched cereal gripper-slip rollout: baseline failure (left), recovery-data success (right). Identical evaluation seed and disturbance.

+22.8 pp	5 / 5	0.919
recovery vs. clean labels	training-seed wins	temporal risk AUROC

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Simulation-only. Runs locally on Apple Silicon or CPU. No CUDA, cloud compute, paid API, or robot hardware required.

Abstract

Interactive imitation learning is motivated by a simple asymmetry: an expert correction at a failure state may be more useful than another successful demonstration. InterveneSim-X turns that intuition into a reproducible, equal-label-budget simulation study that runs on Apple Silicon without CUDA, cloud compute, paid APIs, or robot hardware. We evaluate behavior cloning across four Panda pick-and-place object domains, four deployment disturbances, five independently trained policy seeds, and four additional-label budgets. At the largest matched budget, ordinary behavior cloning on corrective recovery actions achieves 61.4% disturbed success (95% hierarchical bootstrap interval 56.6-65.9), compared with 38.6% (34.5-42.8) for the same number of additional clean labels. The paired gain is 22.8 percentage points and is positive for all five training seeds. A contrastive objective that repels the rejected robot action does not improve performance, demonstrating that more structured supervision is not automatically better. An exploratory temporal risk gate exposes a second distinction: high offline AUROC does not guarantee selective online intervention. A complete threshold sweep finds an operating point with 75.8% assisted disturbed success, 64.8% disturbed interventions, and a 3.1% nominal false-alarm rate. The repository includes the simulator, data schemas, crash-resumable runner, tests, raw episode-level outputs, uncertainty analysis, figures, and matched rollout video.

1. Motivation

Behavior cloning trains on states visited by an expert but deploys on states induced by its own imperfect actions. This distribution mismatch can compound over time. Interactive imitation learning addresses the mismatch by querying or accepting expert control on the learner's state distribution. Yet a practical question often gets lost in algorithmic complexity: if labeling effort is fixed, are corrections actually more valuable than more clean demonstrations?

InterveneSim-X isolates that comparison. The unit of budget is an action label, not a trajectory, and the clean and recovery conditions use exactly the same number of added labels. The goal is not to claim a new state-of-the-art robotics algorithm. It is to build a small experiment whose causal comparison, negative results, and limitations are visible.

2. Experimental design

The environment is robosuite PickPlace with a Panda arm and four object variants: can, milk, bread, and cereal. A scripted expert generates base demonstrations. A baseline MLP policy is then rolled out under object shifts, Gaussian action noise, multi-step action delay, and gripper slip. A privileged supervisor takes over after a disturbance or stalled progress, producing corrective actions at learner-visited states.

Four policy conditions are compared:

- **baseline:** base clean demonstrations only;
- **more_demos:** baseline data plus an action-budgeted sample of additional clean data;
- **recovery_bc:** baseline data plus the same number of corrective recovery actions;
- **contrastive_recovery:** the recovery data plus a margin loss that pushes predictions

away from the robot action rejected at the intervention state.

The policy uses a shared MLP trunk with task- and phase-routed action heads. The 35-dimensional state includes end-effector and object geometry, gripper state, goal geometry, control progress, derived task predicates, and a task one-hot vector. This augmentation resolves phase ambiguity for tall objects while retaining backward compatibility with the released 24-dimensional single-task benchmark.

The primary endpoint is mean success over disturbed rollouts at 4,800 added labels. Training seed is the unit of method-level uncertainty. Five independent training seeds are evaluated on matched episode seeds. A hierarchical bootstrap resamples training seeds and episodes. Exact sign permutations are reported as a small-sample diagnostic; with five seeds, the minimum attainable two-sided value for a same-direction effect is 0.0625.

3. Autonomous-policy results

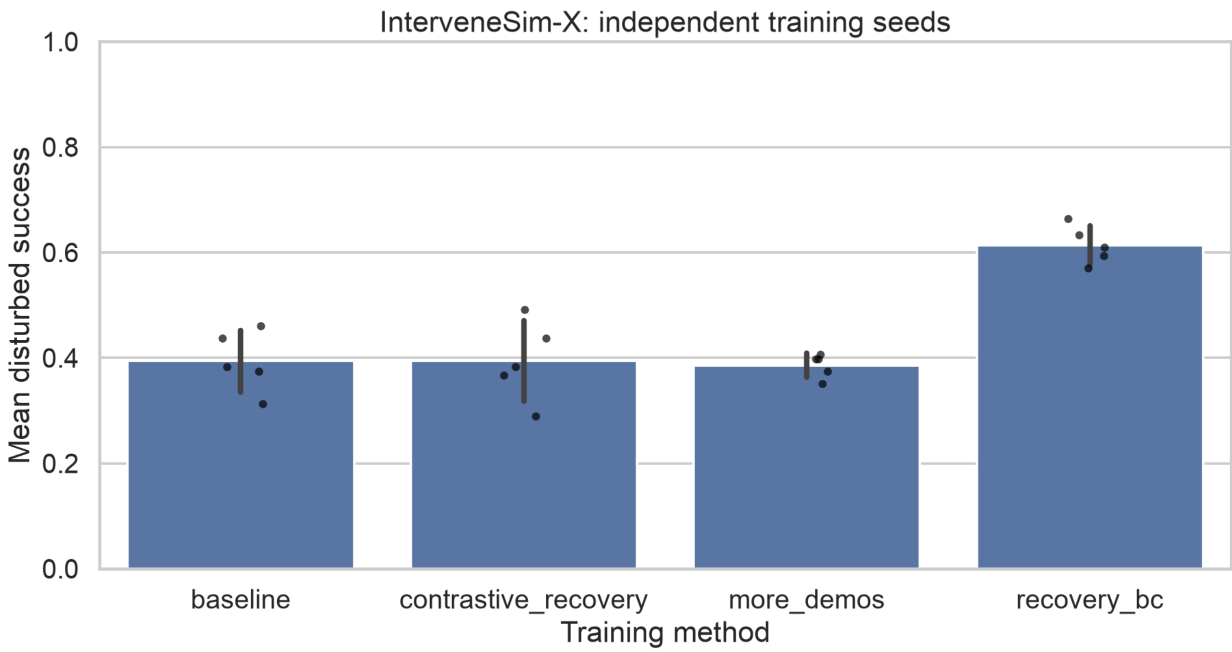

Figure 1. Disturbed success across five independent training seeds. Bars show means and standard deviations; points are individual seeds.

Condition	Success	Std.	Hierarchical 95% CI
Baseline	39.4%	5.8 pp	[33.9, 45.5]
+ Clean labels	38.6%	2.3 pp	[34.5, 42.8]
+ Recovery labels	61.4%	3.6 pp	[56.6, 65.9]
+ Recovery + contrast	39.4%	7.6 pp	[32.5, 46.2]

At the largest matched budget, recovery behavior cloning reaches 61.4% disturbed success, versus 38.6% for additional clean demonstrations and 39.4% for the unaugmented baseline. The recovery-versus-clean difference is +22.8 percentage points with a 4.3-point standard deviation across seeds, and recovery wins for every training seed.

The improvement is disturbance-dependent. Recovery behavior cloning reaches 95.6% under action noise, 60.0% under gripper slip, 60.6% under object shift, and 29.4% under action delay. The matched clean-data condition reaches 76.2%, 9.4%, 50.6%, and 18.1%, respectively. Gripper slip is the clearest example of targeted value: clean labels rarely show how to reacquire a dropped object, while correction data concentrates precisely on that state distribution.

The reference-seed budget curve improves from 48.4% disturbed success at 500 recovery labels to 57.0% at 1,500 and 64.1% at 3,000, then measures 60.9% at 4,800. The last point is not monotonic, which cautions against claiming that every additional correction helps.

4. Negative result: rejected-action contrast

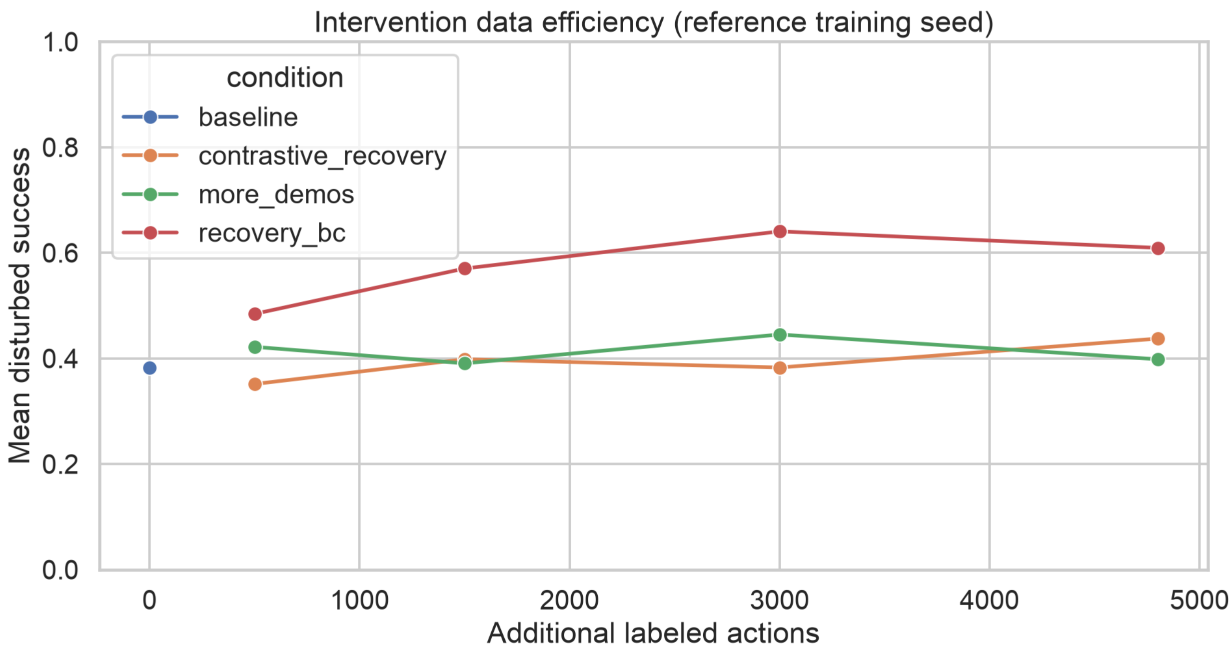

Figure 2. Reference-seed disturbed success across equal added-label budgets. The recovery curve is not monotonic at the final budget.

The dataset records both the expert correction and the baseline action rejected at each intervention state. This enables a contrastive loss that attracts the policy to the expert action and repels it from a sufficiently different rejected action. Despite the additional structure, the contrastive condition obtains only 39.4% disturbed success at 4,800 labels, 22.0 points below ordinary recovery behavior cloning. It loses to ordinary recovery training for all five seeds and also reduces nominal performance.

The likely explanation is optimization interference: the rejected action is not always globally wrong, only wrong in context, and a fixed action-space margin can over-penalize nearby controls that remain useful during other phases. The result argues for learned advantages, state-conditional preferences, or ranking objectives with calibrated weights rather than naive geometric repulsion.

5. Learned help seeking

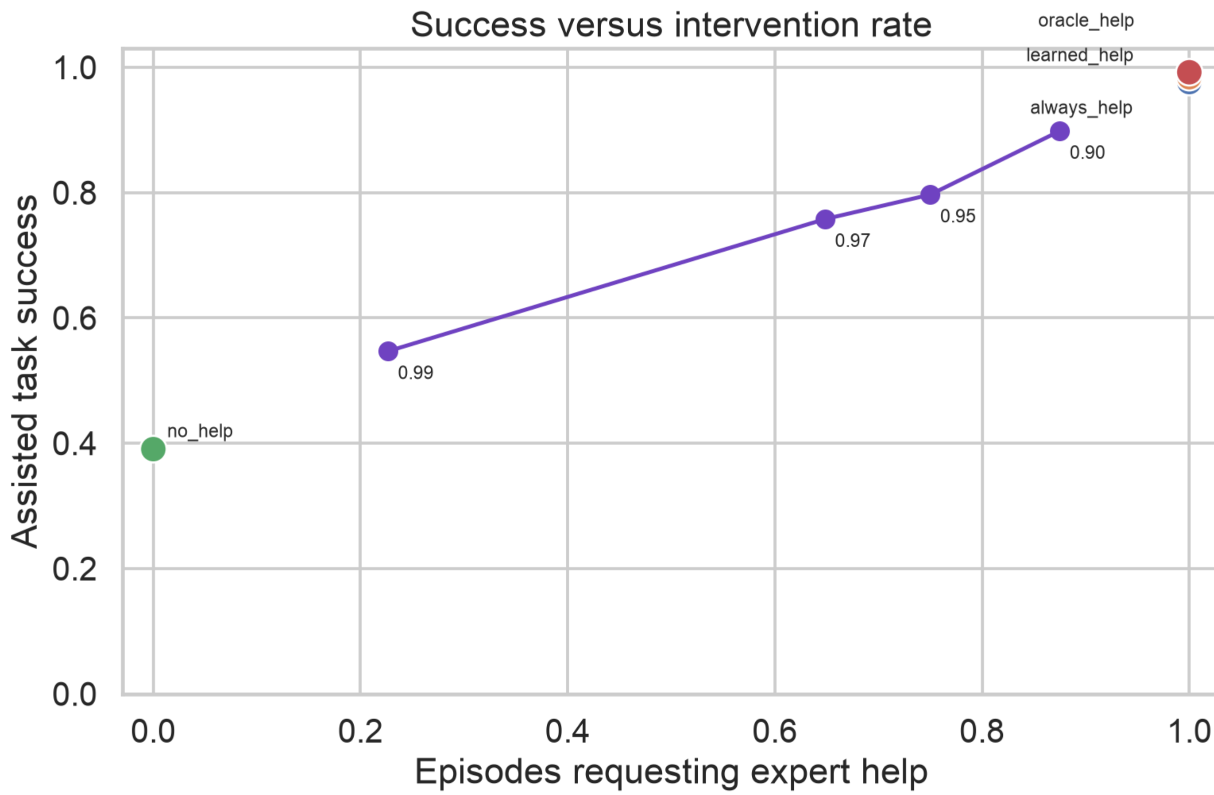

Figure 3. Assisted success versus expert-intervention rate. Purple points show the complete exploratory threshold sweep.

Threshold	Disturbed success	Help on disturbed	Help on nominal
0.90	89.8%	87.5%	28.1%
0.95	79.7%	75.0%	15.6%
0.97	75.8%	64.8%	3.1%
0.99	54.7%	22.7%	0.0%

A static risk classifier was initially trained to predict supervisor intervention within 16 steps. Its offline ranking appeared acceptable, but it requested help at reset on every deployment episode. This is a causal failure: before a randomized disturbance occurs, its state can be identical to a nominal state, so no state-only classifier can infer the future random event.

The corrected exploratory detector combines the current observation with its causal one-step change and predicts intervention within three steps. On episode-disjoint validation data it reaches AUROC 0.919 and average precision 0.527. The validation-selected high-recall threshold remains non-selective online, so a complete post-audit sweep over 0.90, 0.95, 0.97, and 0.99 is reported. Threshold 0.97 yields 75.8% assisted disturbed success while requesting help on 64.8% of disturbed episodes and 3.1% of nominal episodes. This operating point is exploratory, not confirmatory.

The gate experiment demonstrates why offline detector metrics are insufficient. The scientifically relevant object is the deployed success-intervention frontier, including false alarms on nominal trajectories.

6. Reproducibility and boundaries

The complete study uses 13,996 base actions, 5,699 additional clean actions, 7,887 corrective actions with paired rejected actions, and 4,233 risk examples. All experiments run locally with PyTorch MPS or CPU. The runner caches data, checkpoints, and per-condition episode results, so interrupted multi-hour runs resume without recomputing completed work. The public bundle includes resolved configuration, seed-level results, paired comparisons, threshold sweep, figures, and matched video.

The study is state-based simulation with scripted corrections. It does not establish visual robustness, real-world transfer, human correction quality, safety, or equivalence to large vision-language-action models. The recovery supervisor bypasses synthetic actuator corruption after takeover; persistent scene changes remain. These choices define the scope of the recovery claim.

7. Conclusion

Under a controlled action-label budget, corrections gathered on the learner's failure distribution are substantially more useful than additional clean demonstrations in this benchmark. The result replicates across five independently trained policies and four object domains. Two failures are equally important: naive rejected-action contrast hurts, and high offline risk ranking does not by itself produce selective online assistance. Together, these findings make InterveneSim-X a compact platform for studying what to label, how to learn from corrections, and when a deployed robot should ask for help.

References

1. Ross, S., Gordon, G., and Bagnell, D. *A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning.* AISTATS, 2011. https://proceedings.mlr.press/v15/ross11a.html 2. Kelly, M., Sidrane, C., Driggs-Campbell, K., and Kochenderfer, M. J. *HG-DAgger: Interactive Imitation Learning with Human Experts.* ICRA, 2019. https://arxiv.org/abs/1810.02890 3. Hoque, R. et al. *ThriftyDAgger: Budget-Aware Novelty and Risk Gating for Interactive Imitation Learning.* CoRL, 2021. https://arxiv.org/abs/2109.08273 4. Zhu, Y. et al. *robosuite: A Modular Simulation Framework and Benchmark for Robot Learning.* 2020. https://robosuite.ai/